$\therefore$ eff in 1st Construction = eff in 2nd Construction.

eff 1 = eff 2.

As of Pantro eff = $\frac{\text{Work}}{\text{T.N.D/Total time}}$

$\therefore \frac{\text{Work 1}}{\text{Time 1}} = \frac{\text{Work 2}}{\text{Time 2}}$

→ Suppose M_1 men can complete in D_1 days working H_1 hours to complete W_1 work and M_2 men can complete in D_2 days of W_2 work working H_2 hr's per day. Then to find out any value.

$$\frac{\text{Work 1}}{\text{Time 1}} = \frac{\text{Work 2}}{\text{Time 2}}$$

$$\frac{W_1}{M_1 D_1 H_1} = \frac{W_2}{M_2 D_2 H_2}$$

$M_1 D_1 H_1 = M_2 D_2 H_2$ [when works are same]

$$\frac{W_1}{M_1 D_1} = \frac{W_2}{M_2 D_2}$$ [when Hours not given]

Time is 'M' men for 'D' days working 'H' hours a day.

Ex:- 20 Men, 10 Days, 12 hours

1 day time = 12 hrs

1 Men 10 days time = 1 × 10 day × 12 hrs each = 10 × 12

20 Men 10 days time =

= 20 Men × 10 days × 12 hrs

= 20 × 10 × 12.

$\therefore$ Time = Men/Women × Days × Hours

$$\boxed{T = M/W \times D \times H}$$

Ex:6:- 15 men can complete 100 m wall in 20 days working 12 hrs per days. Then how many hrs's per day 45 men should work to complete 500 m wall in 80 days?

Ans:- $M_1 = 15$, $W_1 = 100$, $D_1 = 20$, $H_1 = 12$

$M_2 = 45$, $W_2 = 500$, $D_2 = 80$, $H_2 = ?$

$$\frac{W_1}{M_1 D_1 H_1} = \frac{W_2}{M_2 D_2 H_2} \Rightarrow \frac{100^{1}}{15 \times 20 \times 12_{3}} = \frac{500^{5}}{45_{3} \times 80_{4} \times H_2}$$

$\Rightarrow H_2 = 5$ hrs per day

(or)

By using proportionality Rules:-

W	M	D	H
100	15	20	12
500	45	80	$H_2 = ?$

$$H_2 = 12 \times \frac{500}{100} \times \frac{15}{45} \times \frac{20}{80} = 12 \times 5 \times \frac{1}{3} \times \frac{1}{4} = 5 \text{ hrs}$$

(①) (②) (③)

(A) Compare one by one (W, M & D) with 'H' to get H_2.

① → H W

12 100

H_2 500

↳ here 100 m wall constructed by working 12 hrs's per day.

↳ So, to complete 500 m wall hrs's should be more than h_1 [$h_2 > h_1$ because > 12]

$\therefore h_2 = 12 \times \left[\frac{500}{100}\right]$

means to get $h_2 > h_1$, h_1 is multiplied with greater than 1. It is possible when we consider $\frac{500}{100}$ only. not $\frac{100}{500}$.

(65)

②→ Compare hr's with men

H	M
12	15
H_2	45

↳ 15 men can complete working 12 hr's a day. Same can be done by 45 men in less no. of hr's than 12.

↳ Means H_2 is $< H_1(12)$ for this comparision. and it is possible by multiplying 12 with $\frac{15}{45}$ only not $\frac{45}{15}$ $[12 \times [<1]]$. $\Rightarrow \therefore 12 \times \frac{15}{45}$

③→ Compare Days with Hr's:-

H	D
12	20
H_2	80

↳ A work can be done in 20 days working 12 hr's per day. then, in 80 days the same work to be completed means less no. of hr's per day to be worked. [Means $H_2 = H_1 \times [<1]] \Rightarrow \frac{20}{80} < 1$ not $\frac{80}{20}$

$\therefore \Rightarrow H_2 = 12 \times \frac{20}{80}$

Proportionality Rules:-

from ①:- When D & M are Constant $H \propto W$

from ②:- When D & W are Constant $H \propto \frac{1}{M}$

from ③:- when M & W are Constant $H \propto \frac{1}{D}$

.By Considering all comparisions.

$H_2 = 12 \times \frac{500}{100} \times \frac{15}{45} \times \frac{20}{80} = \boxed{5 \text{ hr's}}$

Pipes & Cisterns:-

↳ The main difference between time & work and Pipes & Cisterns is. In time & work there is no consideration (or) importance of '-ve' work to complete the work. But in Pipes & Cisterns. there must be importance to '-ve' work.

Ex:- In construction of building there no point of destruction. [Time & Work.]

If filling up of tank is the work means there must be emptying (-ve work) through pipes. otherwise there is no use of filling the tank.

↳ In Pipes & Cisterns '+ve' work [filling] with '+ve' sign & '-ve' work [Emptying] with '-ve' sign.

Remaining what concepts we discussed till now in time & work all are applicable & considered. in Pipes & Cisterns too.

Ex:-7:- Pipe 'A' can fill a tank in 36 hr's, Pipe 'B' can fill the same tank in 24 hr's & Pipe 'C' can empty the same tank in 48 hr's. In how many hours the tank full/empty when all the pipes are opened together @ a time?

$A = \frac{1}{36}$, $B = \frac{1}{24}$; $C = \frac{-1}{48}$

Comb eff $= \frac{1}{36} + \frac{1}{24} - \frac{1}{48} = \frac{4+6-3}{144} = \frac{7}{144}$.